UNIT 3 - COMPLETE NOTES

Logarithms

The inverse of powers: what a logarithm is, the four laws that make hard multiplication easy, common logarithms, characteristic and mantissa, and calculating with log tables.

What this unit covers

- **3.1** Introduction - exponential form and logarithmic form
- **3.2** Laws of logarithms - product, quotient, power and change of base
- **3.3** Common logarithms - base 10, characteristic and mantissa
- **3.4** Logarithm tables and antilogarithms - finding a log and reversing it
- **3.5** Calculations with logarithms - products, quotients, powers and roots

How to use these notes

Read one section at a time and work through every worked example with pen and paper before looking at the solution. Yellow boxes hold the rules most often tested in board papers. The last two pages give solved examples, board-style multiple-choice questions and short questions for self-testing.

Exam focus: Board papers regularly ask for (1) converting between exponential and logarithmic form, (2) expanding or joining logarithms using the laws, (3) finding characteristic and mantissa, (4) using log and antilog tables, and (5) evaluating a product, quotient, power or root with logarithms.

Based on *A Textbook of Mathematics for Grade IX*, Khyber Pakhtunkhwa Textbook Board (KPTB), Peshawar - Unit 3: Logarithm, starting at book page 75. Notes prepared by Shah Fahad.

1. From powers to logarithms

You already know powers: $2^3 = 8$, $10^2 = 100$, $3^{-2} = 1/9$. A **logarithm** answers the reverse question: *to what power must the base be raised to get this number?*

> **Definition - Logarithm:** If $a^x = y$, then x is called the **logarithm of y to the base a**, written $\log_a y = x$. Here $a > 0$, $a \neq 1$, and $y > 0$.

$$a^x = y \quad \leftrightarrow \quad \log_a y = x$$

The two statements say exactly the same thing. The first is the **exponential (index) form**; the second is the **logarithmic form**. In $\log_a y$, **a** is the **base** and **y** is the number whose logarithm is taken.

Exponential form	Logarithmic form
$2^5 = 32$	$\log_2 32 = 5$
$10^3 = 1000$	$\log_{10} 1000 = 3$
$3^{-2} = 1/9$	$\log_3 (1/9) = -2$
$5^0 = 1$	$\log_5 1 = 0$
$7^1 = 7$	$\log_7 7 = 1$

Example 1. Write in logarithmic form: (a) $4^3 = 64$ (b) $10^{-2} = 0.01$
Solution: (a) $\log_4 64 = 3$. (b) $\log_{10} 0.01 = -2$.

Example 2. Write in exponential form: (a) $\log_2 16 = 4$ (b) $\log_9 3 = 1/2$
Solution: (a) $2^4 = 16$. (b) $9^{1/2} = 3$.

Example 3. Find x: (a) $\log_2 x = 6$ (b) $\log_x 81 = 2$ (c) $\log_5 25 = x$
Solution: (a) $x = 2^6 = 64$. (b) $x^2 = 81$, so $x = 9$ (base must be positive). (c) $5^x = 25 = 5^2$, so $x = 2$.

> **Conditions - read carefully:** The base must be positive and not equal to 1. The number must be positive: **logarithms of zero and of negative numbers do not exist** in real numbers, because no power of a positive base gives 0 or a negative result.

2. Laws of logarithms - I

Logarithms turn hard operations into easy ones: multiplication becomes addition, division becomes subtraction, and powers become simple multiplication. That is the whole purpose of the laws. Throughout, $a > 0$, $a \neq 1$, and $m, n > 0$.

Law 1 - Product law

$$\log_a (mn) = \log_a m + \log_a n$$

Check: $\log_2 8 + \log_2 4 = 3 + 2 = 5$, and $\log_2 (8 \times 4) = \log_2 32 = 5$. The law works.

Law 2 - Quotient law

$$\log_a (m / n) = \log_a m - \log_a n$$

Check: $\log_{10} 100 - \log_{10} 10 = 2 - 1 = 1$, and $\log_{10} (100 / 10) = \log_{10} 10 = 1$.

Law 3 - Power law

$$\log_a m^n = n \log_a m$$

The exponent comes down in front. It also works for roots, because a root is a fractional power: $\log_a \sqrt{m} = \log_a m^{1/2} = (1/2) \log_a m$.

Example 4. Expand $\log_a (x^3 y / z^2)$ into separate logarithms.

Solution: $= \log_a x^3 + \log_a y - \log_a z^2 =$ **$3 \log_a x + \log_a y - 2 \log_a z$**.

Example 5. Join into a single logarithm: $2 \log x + 3 \log y - \log z$.

Solution: $= \log x^2 + \log y^3 - \log z =$ **$\log (x^2 y^3 / z)$**.

Example 6. Given $\log 2 = 0.3010$ and $\log 3 = 0.4771$ (base 10), find $\log 6$, $\log 12$ and $\log 5$.

Solution: $\log 6 = \log 2 + \log 3 = 0.3010 + 0.4771 =$ **0.7781**. $\log 12 = \log (2^2 \times 3) = 2(0.3010) + 0.4771 =$ **1.0791**. $\log 5 = \log (10 / 2) = 1 - 0.3010 =$ **0.6990**.

> **Direction matters:** Expanding splits one log into many; joining does the reverse. In both directions, deal with powers **last** when expanding and **first** when joining.

3. Laws of logarithms - II

Two values to memorise

$$\log_a 1 = 0 \quad \text{and} \quad \log_a a = 1 \quad \text{for every valid base } a$$

Why? Because $a^0 = 1$ and $a^1 = a$. In general $\log_a a^n = n$, and $a^{\log_a m} = m$ - power and logarithm cancel.

Law 4 - Change of base

$$\log_a m = \log_b m / \log_b a$$

This turns any logarithm into base 10, the base of log tables. Special case (**inverse rule**): $\log_a b = 1 / \log_b a$.

Example 7. Find $\log_2 5$, given log 2 = 0.3010 and log 5 = 0.6990.

Solution: $\log_2 5 = \log 5 / \log 2 = 0.6990 / 0.3010 =$ **2.322** (approx.). Check: $2^{2.322}$ is about 5.

Example 8. Show the quotient law follows from the product and power laws.

Solution: $m / n = m \times n^{-1}$, so $\log_a (m / n) = \log_a m + \log_a n^{-1} = \log_a m - \log_a n$.

Example 9. Given log 2 = 0.3010, evaluate log 16 and log 0.5.

Solution: $\log 16 = \log 2^4 = 4 \times 0.3010 =$ **1.2040**. $\log 0.5 = \log (1 / 2) = \log 1 - \log 2 = 0 - 0.3010 =$ **-0.3010**.

Law	Statement
Product	$\log_a (mn) = \log_a m + \log_a n$
Quotient	$\log_a (m/n) = \log_a m - \log_a n$
Power	$\log_a m^n = n \log_a m$
Change of base	$\log_a m = \log_b m / \log_b a$

Not laws - common traps: log (m + n) is **not** log m + log n, and log m / log n is **not** log (m - n). The laws work on products, quotients and powers **inside** the logarithm - never on sums.

4. Common logarithms, characteristic and mantissa

A logarithm to **base 10** is called a **common logarithm**. When no base is written, base 10 is understood: log 100 = $\log_{10}$ 100 = 2. Common logarithms are the ones printed in log tables.

> **Two parts of a common logarithm:** logarithm = **characteristic** + **mantissa**. The characteristic is the integer part; the mantissa is the decimal part, and it is **always positive** (between 0 and 1).

$$\log 3450 = 3.5378 \quad - \quad \text{characteristic} = 3, \quad \text{mantissa} = 0.5378$$

Finding the characteristic - no table needed

- **Number greater than 1:** characteristic = (digits before the decimal point) - 1. For 3450: 4 digits, so 3. For 47.2: 2 digits, so 1.
- **Number less than 1:** characteristic = -(zeros between the point and the first non-zero digit, plus 1). For 0.0456: 1 zero before the 4, so -2.

The mantissa depends only on the digits

3450, 345, 34.5, 3.45 and 0.345 all have the **same mantissa**, because they share the digits 345 - moving the decimal point changes only the characteristic. That is why one small table serves for every size of number.

Number	As a x 10^n	Characteristic	Mantissa
3450	3.45 x 10^3	3	Same for all: log 3.45
34.5	3.45 x 10^1	1	
3.45	3.45 x 10^0	0	
0.0345	3.45 x 10^{-2}	-2	

Example 10. Write the characteristic of: (a) 927.4 (b) 8.63 (c) 0.00719
Solution: (a) 3 digits before the point, so **2**. (b) 1 digit before the point, so **0**. (c) 2 zeros before the first non-zero digit, so **-3**.

> **Bar notation:** A negative characteristic is written with a bar over it, keeping the mantissa positive. log 0.0456 = -2 + 0.6590, written $\bar{2}.6590$. The bar applies **only** to the integer part - never read it as -2.6590.

5. Log tables and antilogarithms

Finding a logarithm from the table - 4 steps

1. **Characteristic:** count it from the position of the decimal point (Section 4).
2. **Mantissa:** first two significant digits give the row, the third digit gives the column; read the value.
3. **Mean difference:** for a fourth significant digit, add the mean difference from the same row.
4. **Combine:** logarithm = characteristic + mantissa.

Example 11. Find log 63.25.
Solution: Characteristic: 2 digits before the point, so 1. Mantissa: row 63, column 2 gives 0.8007; mean difference for the fourth digit 5 adds 3, giving 0.8010. So log 63.25 = **1.8010**.

Example 12. Find log 0.00456.
Solution: Characteristic: 2 zeros before the 4, so -3. Digits 456 give log 4.56 = 0.6590 (row 45, column 6). So log 0.00456 = $\bar{3}$**.6590**.

Antilogarithm - going backwards

Antilogarithm: If log x = y, then x = antilog y. It reverses the table: the mantissa gives the digits, the characteristic places the decimal point.

1. Use only the **mantissa** in the antilog table: first two digits as row, third as column, plus the mean difference for the fourth digit. You get a 4-digit number.
2. Place the decimal point using the **characteristic**: digits before the point = characteristic + 1. For a bar characteristic, count zeros after the point instead.

Example 13. Find antilog 2.3010 and antilog 1.7782.
Solution: Mantissa 0.3010 gives digits 2000; characteristic 2 means 3 digits before the point: antilog 2.3010 = **200**. Mantissa 0.7782 gives 6000; characteristic 1: antilog 1.7782 = **60**.

Example 14. Find antilog $\bar{2}$.6590.
Solution: Mantissa 0.6590 gives digits 4560. Bar 2 means 1 zero after the point before the first digit: **0.0456**.

Golden check: log and antilog undo each other. If the antilog does not give back your starting number, the decimal point - not the digits - is usually the mistake.

6. Calculations with logarithms

The full method has the same shape every time: **take logs - do the easy operation - take the antilog**. Before calculators, this is how all long multiplication, division, powers and roots were done.

Multiplication - add the logs

Example 15. Evaluate 32.6 x 4.25.
Solution: log 32.6 = 1.5132 and log 4.25 = 0.6284. Sum = 2.1416. Antilog of 2.1416 = **138.5**. Check: 32.6 x 4.25 = 138.55.

Division - subtract the logs

Example 16. Evaluate 456 / 12.3.
Solution: log 456 = 2.6590 and log 12.3 = 1.0899. Difference = 1.5691. Antilog of 1.5691 = **37.07**. Check: 456 / 12.3 = 37.07.

Powers - multiply the log

Example 17. Evaluate $(5.2)^3$.
Solution: log $(5.2)^3$ = 3 x log 5.2 = 3 x 0.7160 = 2.1480. Antilog of 2.1480 = **140.6**. Check: 5.2^3 = 140.608.

Roots - divide the log

Example 18. Evaluate √156.
Solution: log √156 = (1/2) x log 156 = (1/2) x 2.1931 = 1.0966. Antilog of 1.0966 = **12.49**. Check: 12.49^2 is about 156.

> **Roots with a bar characteristic:** Convert first, operate, then rewrite with a positive mantissa. Example - log √0.00456: log 0.00456 = $\bar{3}$.6590 = -3 + 0.6590. Halving gives -1.5 + 0.3295 = -2 + 0.8295 = $\bar{2}$.8295. Antilog ≈ **0.0675**, and 0.0675^2 ≈ 0.00456.

Answers from tables are approximate - usually correct to 3 or 4 significant figures. Write ≈ (approximately) when an answer comes from a log table.

7. Solved examples - mixed practice

Example 19. Solve $\log_3 (x + 2) = 3$ for x.

Solution: In exponential form: $x + 2 = 3^3 = 27$, so **x = 25**.

Example 20. Express $\log_a (\sqrt{x} \times y^2 / z)$ in expanded form.

Solution: $\sqrt{x} = x^{1/2}$, so the expression = $(1/2) \log_a x + 2 \log_a y - \log_a z$.

Example 21. Given log 3 = 0.4771 and log 7 = 0.8451, find log 21, log 49 and log (7/3).
Solution: log 21 = log 3 + log 7 = 0.4771 + 0.8451 = **1.3222**. log 49 = 2 log 7 = **1.6902**. log (7/3) = 0.8451 - 0.4771 = **0.3680**.

Example 22. Find the value of $\log_4 32$.

Solution: Write both numbers as powers of 2: $32 = 2^5$ and $4 = 2^2$. If $4^x = 32$ then $2^{2x} = 2^5$, so 2x = 5 and **x = 5/2 = 2.5**.

Example 23. How many digits are in 2^{20}? Given log 2 = 0.3010.

Solution: $\log 2^{20} = 20 \times 0.3010 = 6.0200$. The characteristic is 6, so the number has 6 + 1 = **7 digits**. Check: $2^{20} = 1{,}048{,}576$.

Example 24. Evaluate $(23.4)^2 / 5.67$ using logarithms, given log 23.4 = 1.3692 and log 5.67 = 0.7536.
Solution: log of the expression = 2(1.3692) - 0.7536 = 2.7384 - 0.7536 = 1.9848. Antilog of 1.9848 ≈ **96.6**. Check: $23.4^2 = 547.56$, and 547.56 / 5.67 ≈ 96.6.

Common mistakes to avoid

- Writing log (a + b) = log a + log b. **Wrong** - the product law needs a product inside, not a sum.
- Forgetting the conditions: log of 0 or of a negative number has no real value.
- Reading bar 2 point 6590 as -2.6590. The bar covers the characteristic only; the mantissa stays positive.
- Using the mantissa to place the decimal point in an antilog. Digits come from the mantissa; **position** comes from the characteristic.
- Mixing bases in one step: the laws need the **same base** on both sides.

8. Test yourself - MCQs and short questions

Multiple-choice questions

1. $\log_2 32$ equals: (a) 4 (b) 5 (c) 6 (d) 16
   Answer: (b) - $2^5 = 32$.
2. The logarithmic form of $3^4 = 81$ is: (a) $\log_4 81 = 3$ (b) $\log_3 4 = 81$ (c) $\log_3 81 = 4$ (d) $\log_{81} 3 = 4$
   Answer: (c)
3. $\log_a (mn)$ equals: (a) $\log_a m \times \log_a n$ (b) $\log_a m + \log_a n$ (c) $\log_a m - \log_a n$ (d) $(\log_a m)^n$
   Answer: (b) - product law.
4. log 1000 equals: (a) 10 (b) 100 (c) 3 (d) 1/3
   Answer: (c) - $10^3 = 1000$.
5. The characteristic of log 0.0073 is: (a) 2 (b) -2 (c) 3 (d) -3
   Answer: (d) - 2 zeros before the first non-zero digit, so -(2 + 1) = -3.
6. If log 2 = 0.3010, then log 8 equals: (a) 0.6020 (b) 0.9030 (c) 0.3010 (d) 2.4080
   Answer: (b) - log 8 = 3 log 2 = 0.9030.
7. $\log_5 1$ equals: (a) 1 (b) 5 (c) 0 (d) undefined
   Answer: (c) - $\log_a 1 = 0$ for every base.
8. Antilog of 0.4771 is: (a) 2 (b) 3 (c) 30 (d) 0.3
   Answer: (b) - log 3 = 0.4771.

Short questions for written practice

1. Convert to logarithmic form: (a) $2^6 = 64$ (b) $10^{-3} = 0.001$. Convert to exponential form: (c) $\log_4 256 = 4$.
2. Expand completely: $\log_a (p^2 \sqrt{q} / r^3)$.
3. Write as a single logarithm: (1/2) log x + 2 log y - 3 log z. (Answer: log ($\sqrt{x}\, y^2 / z^3$))
4. Write the characteristic of: (a) 5820 (b) 63.9 (c) 0.052 (d) 0.00084.
5. Given log 2 = 0.3010 and log 3 = 0.4771, find log 18 and log 1.5. (Answers: 1.2552; 0.1761)
6. Use change of base to find $\log_3 10$, given log 3 = 0.4771. (Answer: 1 / 0.4771 ≈ 2.096)

One-page revision: $a^x = y \leftrightarrow \log_a y = x$. Laws: log (mn) = log m + log n; log (m/n) = log m - log n; log m^n = n log m; change of base: $\log_a m = \log_b m / \log_b a$. Also $\log_a 1 = 0$ and $\log_a a = 1$. Common log = base 10; logarithm = characteristic (from the decimal point) + mantissa (from the digits, always positive). Calculations: multiply - add logs; divide - subtract; power - multiply; root - divide; then take the antilog.

End of Unit 3 notes. Next unit: Unit 4 - Algebraic Expressions and Algebraic Formulas. Source book: A Textbook of Mathematics for Grade IX, KPTB Peshawar.